Database Design and Implementation
Specifications for the Database Design and Implementation Project – Part B

1. Implementing your proposal using Oracle

Use Oracle to implement your proposed database.

- Write a script to create the database schema.
- Write a script to populate the database tables. The number of rows in each table can be kept small through careful planning. The only requirement is that every row-producing query that you implement must result in at least two rows of query results.
- Write a script that contains one query (per table) of the form: **`SELECT * FROM`** ***table;*** in order to print out your database.
- Write a script with the **SQL queries**. Include the following for each query:
  - A comment line stating the feature(s) the query demonstrates (e.g. *Q2 – correlated subquery*).
  - A comment line describing the query in English.
  - The SQL code for the query.
- Using a free SQL cloud-based environment or a similar DB engine, test and run the scripts mentioned above.

2. Submit a final report

The final report must be double-sided and must contain the following items in the order in which they are listed below:

1. A cover page that includes:
   - Project title
   - Team member names
   - Course (section), semester, and year
2. The database requirements specifications (in proper English grammar, spelling, etc.).
3. The ER diagram.
4. The rudimentary relational schema which resulted from converting your ER.
5. The BCNF relational schema if your schema in (4) above was not already in BCNF.
6. A printout of the different scripts, including your queries.
7. The printout that you obtained from running your scripts, including your queries.
8. Electronic versions of the material within the final report and submit it via Blackboard, using the final deliverables link.

Query types

At a minimum, your queries must demonstrate the features listed below. *You may of course demonstrate more than one feature in any one query* and thus end up having to write fewer, but more interesting, queries.

1. A join involving at least four relations.
2. A self-join.
3. UNION, INTERSECT, and/or MINUS.
4. SUM, AVG, MAX, and/or MIN.
5. GROUP BY, HAVING, and ORDER BY; all appearing in the same query.
6. Either a correlated subquery or a non-correlated subquery, clearly specify which type of subquery you used.

Notes:

If a query (or feature) doesn't work, write a comment next to that query or feature. You may get partial credit for a serious attempt. On the other hand, if your query doesn't work and I find that out by inspecting your results, then credit will be given.

Submission format for the FINAL report

1. **Team submission**

- The **printed material** (***double-sided***) in the order specified above.
- The original proposal with my comments on it.
- Electronic versions of the material submitted in the final report and submit it via Blackboard, under assignments.

2. Individual submission

Fill out and then print or upload the "Team-Skills Survey Form".

College of Computing **Team-Skills Survey Form**

Course number (section): _____(____) Semester/Year:_____/_____ Name: _____

Why this survey is important
In your career, you will be asked periodically to evaluate your coworkers. This survey should help you start thinking of and practicing for, that task. The School is extremely interested in developing students' team skills. This survey will give the School important feedback regarding what it needs to emphasize in the future.

Legend
SA=Strongly Agree
A=Agree
D=Disagree
SD=Strongly Disagree
NA=Not Applicable or No Opinion

		…Your Team Members' Names …				
		#1 Dave	**#2**	**#3**	**#4**	**#5**
1	**This team member contributed to group discussions.** *Issues to consider include:* *– Participated constructively in meetings.* *– Contributed good ideas to the project.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
2	**This team member was dependable.** *Issues to consider include:* *– Worked around his/her schedule in order to set up meeting times.* *– Attended all meetings.* *– Delivered assigned tasks professionally and on time.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
3	**This team member showed initiative.** *Issues to consider include:* *– Acted to move forward rather than waited to be told what to do.* *– Was essential in the formulation of new ideas and solutions.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
4	**This team member was easy to work with.** *Issues to consider include:* *– Was objective in evaluating other members' ideas.* *– Accepted critique of his ideas graciously.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
5	**This team member contributed fair share of work.** *Issues to consider include:* *– Did at least as much work as any other member.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
6	**This team member communicated effectively in writing.** *Issues to consider include:* *– Expressed ideas clearly in writing (reports, emails, blogs, chats, etc.).*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
7	**This team member communicated well when speaking.** *Issues to consider include:* *– Expressed ideas clearly during discussions and presentations.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
8	**This team member behaved in an ethical manner.** *Issues to consider include:* *– Is trustworthy.* *– Understands and upholds the standards of academic honesty.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*
9	**Given the choice, I would work with this team member again** *Issues to consider include:* *– Should another opportunity arise in this (or in another) class, I'll be willing to work again with this team member.* *– I'll consider myself fortunate if this team member and I are hired in the same company after graduation.* *– If I were working as a team leader in a company, I'll be glad to hire this member in my team.*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*	*SA A D SD N*

Additional Comments*: *Please feel free to use the back of this form or attach another sheet of paper if you wish.